TFPT v2.9: Admissible Vacua, Minimal Carrier, and Electromagnetic Stationarity

A Carrier-First Primary Form

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Abstract

This document rewrites TFPT v2.8 into a carrier-first primary form. The rigid core is the Minimal Carrier Theorem: admissibility on the orientable double cover together with the seam involution forces the minimal carrier $E_3 \oplus E_2$. From this rigid package follow the global Standard Model group $S(U(3) \times U(2))$, one chiral family $S^+ = \Lambda^{\text{even}} E$, the carrier norm $\mathrm{Tr}_E\, Y^2 = 5/6$, and the linear and cubic trace cancellations. On the canonical occupancy branch, the admissible chiral multiplicity is

$$\Omega_{\text{adm}} = N_{\text{fam}} \dim S^+ = 3 \times 16 = 48.$$

The abelian coefficient takes the carrier form

$$b_1(N_{\text{fam}}, N_H) = \frac{4}{3} N_{\text{fam}} + \frac{1}{10} N_H,$$

so the canonical branch $(N_{\text{fam}}, N_H) = (3, 1)$ gives $b_1 = 41/10$. The electromagnetic stationarity equation can then be written entirely in terms of the carrier packet

$$\mathcal{C}_{\min} = (E_3 \oplus E_2,\ Y,\ S^+,\ \gamma,\ B),$$

with $\gamma = \mathrm{Tr}_E\, Y^2 = 5/6$ and $B = \operatorname{rank} E_3 / \operatorname{rank} E_2 = 3/2$. Seam descent for c_3, the induced Einstein channel, the E_8 reinterpretation, Higgs selection, and discrete phase grammar are retained as structural bridges or dynamical completion targets, but no longer presented as part of the hard carrier core.

Status boundary for the v2.9 rewrite

1. **Hard carrier statements**
   - minimal carrier $E_3 \oplus E_2$,
   - global Standard Model group $S(U(3) \times U(2)) \cong (SU(3) \times SU(2) \times U(1)_Y)/\mathbb{Z}_6$,
   - $S^+ = \Lambda^{\text{even}} E$ as one chiral family with ν^c,
   - carrier invariants $\dim S^+ = 16$, $\gamma = \mathrm{Tr}_E\, Y^2 = 5/6$, and $B = \operatorname{rank} E_3 / \operatorname{rank} E_2 = 3/2$,
   - linear and cubic trace cancellation,
   - the general abelian index formula $b_1(N_{\text{fam}}, N_H) = \frac{4}{3} N_{\text{fam}} + \frac{1}{10} N_H$.
2. **Structural bridges**
   - three families from the three distinguished cycles C_1, C_2, C_T,
   - $\Omega_{\text{adm}} = N_{\text{fam}} \dim S^+$ and $\Omega_{\text{adm}} = 48$ on the canonical branch,
   - $\delta_{\text{top}} = \Omega_{\text{adm}}\, c_3^4$ as occupancy transgression,

- $\delta_2 = B\gamma\,\Omega_{\mathrm{adm}}^2 c_3^8$ and therefore $B\gamma = 5/4$ on the minimal carrier,
- the E_8 rereading $\lambda_{E8} = \gamma/\ln(248/60)$.

3. Dynamical completions

- seam descent for c_3,
- the relative-a_2 / induced-Einstein channel,
- the light seam-even Higgs candidate in E_2,
- the physical realization of the $\mathbb{Z}_6 \rtimes \mathbb{Z}_2$ phase grammar.

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1 Overview and Claim Map

The v2.9 rewrite should be read from admissibility to carrier, from carrier to occupancy, and only then from occupancy to closure and phenomenology. The master line is therefore

$$(\widetilde{M}, \Gamma, \tau) \Rightarrow E_3 \oplus E_2 \Rightarrow Y,\ S^+,\ \gamma,\ B \Rightarrow b_1,\ \Omega_{\mathrm{adm}} \Rightarrow \mathrm{CFE} \Rightarrow \alpha.$$

The main bridge branches are secondary to this line:

$$\Omega_{\mathrm{adm}} \Rightarrow \delta_{\mathrm{top}},\ \delta_2, \qquad \gamma \Rightarrow \lambda_{E8}, \qquad \Delta a_2^{\mathrm{rel}} \Rightarrow \bar{M}_{\mathrm{Pl}}.$$

The center of the architecture is therefore not the number 48 by itself, but the rigid minimal carrier package that later induces 48 on the canonical occupancy branch.

Claim	Status	Depends on	First derived in section
Minimal carrier $E_3 \oplus E_2$	hard	admissibility on $\widetilde{M}$ and seam involution τ	3
Global Standard Model group	hard	minimal carrier split	3
One chiral family S^+	hard	minimal carrier and half-spinor construction	4
General abelian index $b_1(N_{\mathrm{fam}}, N_H)$	hard	carrier traces plus scalar multiplicity	4
Canonical occupancy $\Omega_{\mathrm{adm}} = 48$	structural bridge	family-index bridge and $\dim S^+ = 16$	5
Carrier form of the CFE	structural bridge	c_3, Ω_{adm}, b_1, B, γ	6
Induced Einstein channel	dynamical target	relative spectral completion and canonical occupancy	8
E_8 rereading	downstream reinterpretation	carrier norm γ	9

Table 1: Claim map for the carrier-first v2.9 presentation. Appendix A records a compact v2.8→v2.9 dictionary.

2 Admissibility and Seam Data

The ambient v2.9 datum may be packaged as

$$\mathfrak{T}_{2.9} = (\widetilde{M} \to M,\ \Gamma,\ \tau,\ \chi),$$

but the hard carrier core uses only $(\widetilde{M}, \Gamma, \tau)$. The field χ belongs to the absolute-scale and gravity channels discussed later in Section 8.

2.1 Seam data recalled from v2.8

The seam information needed by the carrier rewrite is already present in v2.8. In the minimal gluing class one has

$$U_\Gamma(\theta) = e^{i\theta}, \qquad \mathrm{SF}(U_\Gamma) = 1, \qquad \Delta_\Gamma = 2\pi.$$

On the orientable double cover the distinguished cycles are

$$C_1, \qquad C_2, \qquad C_T,$$

where C_T is the seam cycle in the glued description and corresponds to $\Gamma_+ \cup \Gamma_-$ in the cut description. These data should appear near the front of the paper because Sections 3–6 all depend on them.

What χ does and does not do

χ belongs to the absolute-scale and gravitational channels. It does *not* enter the hard derivation of the minimal carrier, the global Standard Model group, the one-family half-spinor, or the carrier norm $\mathrm{Tr}_E\, Y^2 = 5/6$. Only after the carrier packet has been fixed does χ enter the relative spectral stationarity problem and the induced-gravity target.

3 The Minimal Carrier Theorem

3.1 Uniqueness of the $3+2$ split

Theorem 3.1 (Minimal carrier theorem)**.** *Let $E = E_b \oplus E_s$ be a split carrier with traceless diagonal generator*

$$Y_{b,s} = \mathrm{diag}\left(-\frac{1}{b}, \ldots, -\frac{1}{b}, \frac{1}{s}, \ldots, \frac{1}{s}\right).$$

If one simultaneously requires

$$\dim \Lambda^{\mathrm{even}} E = 16 \qquad \textit{and} \qquad \mathrm{Tr}_E\, Y_{b,s}^2 = \frac{5}{6},$$

then necessarily $\{b, s\} = \{3, 2\}$.

Proof. For a rank-g carrier with $g = b + s$, the positive half-spinor has dimension

$$\dim \Lambda^{\mathrm{even}} E = 2^{g-1}.$$

Demanding one chiral family gives $2^{g-1} = 16$, hence $g = 5$ and therefore $b + s = 5$.

For the traceless generator above one has

$$\mathrm{Tr}_E\, Y_{b,s}^2 = b\left(\frac{1}{b^2}\right) + s\left(\frac{1}{s^2}\right) = \frac{1}{b} + \frac{1}{s} = \frac{b+s}{bs} = \frac{5}{bs}.$$

Imposing $\mathrm{Tr}_E\, Y_{b,s}^2 = 5/6$ yields $bs = 6$. Together with $b + s = 5$, this forces $(b, s) = (3, 2)$ up to order. □

Hence from now on we set

$$E = E_3 \oplus E_2, \qquad P_3 := \frac{1-\tau}{2}, \qquad P_2 := \frac{1+\tau}{2}.$$

Proposition 3.2 (Hypercharge as an affine form of the seam involution)**.** For the minimal carrier,

$$Y = -\frac{1}{3}P_3 + \frac{1}{2}P_2 = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right),$$

and equivalently

$$Y = \frac{1}{12}\mathbf{1} + \frac{5}{12}\tau.$$

Proof. Using $P_3 = (1-\tau)/2$ and $P_2 = (1+\tau)/2$ gives

$$-\frac{1}{3}P_3 + \frac{1}{2}P_2 = -\frac{1}{6}(1-\tau) + \frac{1}{4}(1+\tau) = \frac{1}{12}\mathbf{1} + \frac{5}{12}\tau.$$

□

Theorem 3.3 (Global Standard Model theorem)**.** *The determinant-preserving connected stabilizer of the minimal carrier split is*

$$\mathrm{Aut}_0(E_3 \oplus E_2, \det = 1) = S(U(3) \times U(2)) \cong \frac{SU(3) \times SU(2) \times U(1)_Y}{\mathbb{Z}_6}.$$

Proof. Automorphisms preserving the split and the determinant condition act independently on E_3 and E_2 with unit determinant on the combined carrier. This gives $S(U(3) \times U(2))$. The covering map from $SU(3) \times SU(2) \times U(1)_Y$ to $S(U(3) \times U(2))$ has the canonical $\mathbb{Z}_6$ subgroup as kernel, yielding the stated global Standard Model form. □

Set

$$S^+ := \Lambda^{\mathrm{even}}(E_3 \oplus E_2).$$

Proposition 3.4 (Carrier invariants)**.** For the minimal carrier,

$$\gamma := \mathrm{Tr}_E\, Y^2 = \frac{5}{6}, \qquad B := \frac{\mathrm{rank}\, E_3}{\mathrm{rank}\, E_2} = \frac{3}{2}, \qquad \dim S^+ = 16.$$

Proof. The first identity is

$$\mathrm{Tr}_E\, Y^2 = 3\left(\frac{1}{9}\right) + 2\left(\frac{1}{4}\right) = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}.$$

The second is immediate from the split ranks. The third follows from

$$\dim S^+ = \dim \Lambda^{\mathrm{even}} E = 2^{5-1} = 16.$$

□

Definition 3.5 (Minimal carrier packet)**.** The rigid carrier packet of v2.9 is

$$\mathcal{C}_{\min} := (E_3 \oplus E_2,\ Y,\ S^+,\ \gamma,\ B).$$

4 Carrier Matter, Trace Identities, and the Abelian Index

4.1 One chiral family from the positive half-spinor

Theorem 4.1 (Carrier matter theorem)**.** *For the minimal carrier $E = E_3 \oplus E_2$, the positive half-spinor*

$$S^+ := \Lambda^{\mathrm{even}}(E_3 \oplus E_2)$$

decomposes as

$$(1,1)_0 \oplus (3,2)_{1/6} \oplus (\bar{3},1)_{-2/3} \oplus (\bar{3},1)_{1/3} \oplus (1,2)_{-1/2} \oplus (1,1)_1.$$

Thus S^+ carries exactly one Standard Model family with a right-handed neutrino:

$$\nu^c,\ Q,\ u^c,\ d^c,\ L,\ e^c.$$

Proof. The even exterior powers split as

$$\Lambda^{\text{even}} E = \Lambda^0 E \oplus \Lambda^2 E \oplus \Lambda^4 E.$$

Now

$$\Lambda^0 E = (1,1)_0,$$
$$\Lambda^2 E = \Lambda^2 E_3 \oplus (E_3 \otimes E_2) \oplus \Lambda^2 E_2 = (\bar{3},1)_{-2/3} \oplus (3,2)_{1/6} \oplus (1,1)_1,$$

and $\Lambda^4 E \cong E^*$ because $\det E$ is trivial in the $S(U(3) \times U(2))$ form, so

$$\Lambda^4 E \cong (\bar{3},1)_{1/3} \oplus (1,2)_{-1/2}.$$

Collecting the terms gives the stated multiplets. □

4.2 Trace identities

Theorem 4.2 (General half-spinor quadratic trace identity)**.** *Let Y be a traceless diagonal generator on a rank-g carrier E. Then on the positive half-spinor*

$$S^+ = \Lambda^{\text{even}} E$$

one has

$$\operatorname{Tr}_{S^+} Y^2 = 2^{g-3} \operatorname{Tr}_E Y^2.$$

Proof. Write the diagonal weights of Y as q_i with $\sum_i q_i = 0$. Then

$$\operatorname{Tr}_{S^+} Y^2 = \sum_{S \text{ even}} \left(\sum_{i \in S} q_i \right)^2.$$

Each q_i^2 appears in 2^{g-2} even subsets and each pair $q_i q_j$ appears in 2^{g-3} even subsets. Using

$$\left(\sum_i q_i \right)^2 = 0$$

shows that the cross terms cancel half the self-term contribution, leaving

$$\operatorname{Tr}_{S^+} Y^2 = 2^{g-3} \sum_i q_i^2 = 2^{g-3} \operatorname{Tr}_E Y^2.$$

□

Corollary 4.3 (Minimal carrier quadratic trace)**.** *For the minimal carrier,*

$$\operatorname{Tr}_{S^+} Y^2 = 2^{5-3} \operatorname{Tr}_E Y^2 = 4 \cdot \frac{5}{6} = \frac{10}{3}.$$

Theorem 4.4 (Linear and cubic trace cancellation)**.** *On the one-family carrier spectrum S^+ one has*

$$\operatorname{Tr}_{S^+} Y = 0, \qquad \operatorname{Tr}_{S^+} Y^3 = 0.$$

Proof. Using the explicit family decomposition above, the hypercharges are

$$0, \quad \frac{1}{6}, \quad -\frac{2}{3}, \quad \frac{1}{3}, \quad -\frac{1}{2}, \quad 1,$$

with multiplicities $1, 6, 3, 3, 2, 1$. Hence

$$\operatorname{Tr}_{S^+} Y = 6 \cdot \frac{1}{6} + 3 \cdot \left(-\frac{2}{3}\right) + 3 \cdot \frac{1}{3} + 2 \cdot \left(-\frac{1}{2}\right) + 1 = 0,$$

and

$$\operatorname{Tr}_{S^+} Y^3 = 6 \cdot \left(\frac{1}{6}\right)^3 + 3 \cdot \left(-\frac{2}{3}\right)^3 + 3 \cdot \left(\frac{1}{3}\right)^3 + 2 \cdot \left(-\frac{1}{2}\right)^3 + 1^3 = 0.$$

□

4.3 General abelian index

Theorem 4.5 (General carrier formula for the abelian coefficient)**.** *For a branch with N_{fam} copies of the carrier family and N_H light weak doublets from E_2, one has*

$$b_Y(N_{\text{fam}}, N_H) := \frac{2}{3} N_{\text{fam}} \operatorname{Tr}_{S^+} Y^2 + \frac{1}{3} N_H \operatorname{Tr}_{E_2} Y^2.$$

For the minimal carrier this becomes

$$b_Y(N_{\text{fam}}, N_H) = \frac{2}{3} N_{\text{fam}} \cdot \frac{10}{3} + \frac{1}{3} N_H \cdot \frac{1}{2} = \frac{20}{9} N_{\text{fam}} + \frac{1}{6} N_H.$$

Therefore

$$b_1(N_{\text{fam}}, N_H) = \frac{3}{5}\, b_Y(N_{\text{fam}}, N_H) = \frac{4}{3} N_{\text{fam}} + \frac{1}{10} N_H.$$

Corollary 4.6 (Canonical carrier branch)**.** *For the canonical branch $(N_{\text{fam}}, N_H) = (3, 1)$,*

$$b_Y = \frac{41}{6}, \qquad b_1 = \frac{41}{10}.$$

Remark. The symbol ν^c is already part of the hard packet S^+. Removing it does not define a different carrier; it is an external truncation of the family packet and is treated as such only in Section 7.

5 Boundary Multiplicity and Admissible Occupancy

5.1 Family-index bridge

Conjecture 5.1 (Family-index bridge)**.** Let each distinguished cycle C_i carry an integer family index $I_{\text{fam}}(C_i)$. Then the v2.8 block coefficient may be organized as

$$k_B = \frac{1}{2} \sum_{i=1}^{3} I_{\text{fam}}(C_i)\, I_1(B).$$

Combined with the v2.8 block formula

$$k_B = \frac{3}{2}\, I_1(B),$$

this gives

$$\sum_{i=1}^{3} I_{\text{fam}}(C_i) = 3.$$

The equal-one assignment

$$I_{\text{fam}}(C_1) = I_{\text{fam}}(C_2) = I_{\text{fam}}(C_T) = 1$$

is the minimal symmetric branch and yields

$$N_{\text{fam}} = 3.$$

5.2 Admissible occupancy and topological surcharge

Definition 5.2 (Admissible occupancy)**.** The admissible chiral occupancy is

$$\Omega_{\mathrm{adm}} := \mathrm{Tr}_{\mathcal{H}_{\mathrm{ch}}} \mathbf{1} = N_{\mathrm{fam}} \dim S^{+}.$$

On the canonical branch one therefore has

$$\Omega_{\mathrm{adm}} = 3 \times 16 = 48.$$

Conjecture 5.3 (Occupancy transgression)**.** The topological surcharge should be written as

$$\delta_{\mathrm{top}} = \Omega_{\mathrm{adm}}\, c_3^4.$$

For the canonical branch this becomes

$$\delta_{\mathrm{top}} = 48 c_3^4 = \frac{3}{256\pi^4} = 3 \cdot 16\, c_3^4.$$

In this reading the factor 3 comes from the weighted boundary geometry while the factor 16 comes from the minimal carrier packet itself.

5.3 Seam descent normalization of c_3

The normalization of c_3 is best presented in two stages.

Stage 1: hard seam geometry. On the spin lift of the minimal seam loop one has

$$\widetilde{U}_{\Gamma}(2\pi) = -1, \qquad \widetilde{U}_{\Gamma}(4\pi) = 1.$$

So the spin-lifted loop closes only after 4π.

Stage 2: conditional axionic identification. If the admissible carrier modulus is identified with the axionic angle variable, write

$$a = \frac{\theta}{4\pi}.$$

Then

$$\mathcal{L}_{\theta} = \frac{\theta}{32\pi^2} F\widetilde{F} = \frac{1}{8\pi} a F\widetilde{F},$$

hence

$$c_3 = \frac{1}{8\pi}.$$

The geometry and the final normalization are thus explicitly separated: the 4π closure is geometric, while the identification of the carrier modulus with the axionic angle remains the extra conditional step.

6 Electromagnetic Stationarity and the Carrier CFE

6.1 Carrier form of the stationary equation

The electromagnetic closure channel should first be written on a general admissible branch. Define

$$\varphi_0(\alpha) = \frac{1}{6\pi} + \Omega_{\mathrm{adm}}\, c_3^4 e^{-2\alpha} + B\gamma\, \Omega_{\mathrm{adm}}^2 c_3^8 e^{-4\alpha} + \cdots .$$

Then the carrier form of the CFE is

$$\alpha^3 - 2c_3^3\alpha^2 - 8c_3^6\, b_1(N_{\mathrm{fam}}, N_H) \ln\!\big(\varphi_0(\alpha)^{-1}\big) = 0.$$

This is the preferred v2.9 presentation because the coefficients are now grouped by carrier data rather than as parallel numerical motifs.

Proposition 6.1 (Canonical carrier branch)**.** For

$$(N_{\text{fam}}, N_H, \nu^c) = (3, 1, \text{present}),$$

one has

$$\Omega_{\text{adm}} = 48, \qquad b_1 = \frac{41}{10}, \qquad B\gamma = \frac{3}{2} \cdot \frac{5}{6} = \frac{5}{4}.$$

Hence

$$\varphi_0(\alpha) = \frac{1}{6\pi} + 48 c_3^4 e^{-2\alpha} + \frac{5}{4}\, 48^2 c_3^8 e^{-4\alpha} + \cdots .$$

Remark. The identity

$$\frac{5}{4} = B\gamma$$

is the compressed carrier reading of the v2.8 two-defect weight.

Using

$$c_3 = \frac{1}{8\pi}, \qquad \varphi_{\text{tree}} = \frac{1}{6\pi}, \qquad \delta_{\text{top}} = 48 c_3^4,$$

the positive stationary root remains

$$\alpha^{-1}(0) = 137.035\,999\,2.$$

The point of v2.9 is therefore not to move the number, but to reduce the number of primitive inputs behind it.

Quantity	Carrier meaning	Value
c_3	canonical seam-descent normalization	$\frac{1}{8\pi}$
Ω_{adm}	admissible chiral occupancy	48
γ	carrier norm $\mathrm{Tr}_E\, Y^2$	$\frac{5}{6}$
B	carrier rank ratio $\operatorname{rank} E_3 / \operatorname{rank} E_2$	$\frac{3}{2}$
$B\gamma$	compressed two-defect weight	$\frac{5}{4}$
b_1	canonical abelian coefficient	$\frac{41}{10}$
$\alpha^{-1}(0)$	positive root of the carrier CFE	137.035 999 2

Table 2: Numerical summary of the canonical carrier branch.

6.2 Algebraic corollaries of the carrier kernel

Evaluated on the stationary root of the canonical branch, several previously separate formulas become algebraic corollaries of the same carrier package:

$$g_{a\gamma\gamma} = -4c_3 = -\frac{1}{2\pi}, \qquad \beta_{\text{rad}} = \frac{\varphi_0}{4\pi},$$

$$\lambda_C = \sqrt{\varphi_0(1-\varphi_0)}, \qquad \sin^2\theta_{13} = \varphi_0 e^{-\gamma}.$$

The structural gain is not that these quantities are numerically new, but that they are now visibly downstream of $\mathcal{C}_{\min}$ and the occupancy branch.

6.3 Conditional χ-channel corollaries

If the χ channel and the induced Einstein channel are successfully completed, then the inflationary block can be written as

$$\frac{M}{\chi_0} = \frac{2}{\sqrt{\pi}}\, c_3^4, \qquad \frac{M}{\bar{M}_{\text{Pl}}} = \sqrt{8\pi}\, c_3^4,$$

and therefore

$$A_s = \frac{N^2}{24\pi^2}\left(\frac{M}{\bar{M}_{\rm Pl}}\right)^2 = \frac{N^2}{3\pi}c_3^8.$$

For $N = 56$ this yields

$$A_s = 2.090\,191\,364 \times 10^{-9}.$$

These results are kept separate from the hard carrier corollaries because they depend on the success of the relative-χ and gravity channels.

7 Discrete Exclusion Structure

The carrier-centered rewrite is strongest when it excludes wrong branches, not when it celebrates an isolated number. The relevant deformations fall into three logical classes.

Logical type	Deformation	Carrier/CFE consequence	Result
Boundary multiplicity	$N_{\rm fam} = 2$	$\Omega_{\rm adm} = 32$ and $b_1 = \frac{83}{30}$ in the carrier CFE	$\alpha^{-1}(0) = 156.094\,045\,5$
Boundary multiplicity	$N_{\rm fam} = 4$	$\Omega_{\rm adm} = 64$ and $b_1 = \frac{163}{30}$ in the carrier CFE	$\alpha^{-1}(0) = 124.835\,054\,8$
Scalar content	$N_H = 2$	$b_1 = \frac{21}{5}$ with the same carrier family packet	$\alpha^{-1}(0) = 135.945\,991\,7$
External truncation	no ν^c	the family packet is externally reduced from 16 to 15 states, so $\Omega_{\rm adm} = 45$ and the hard carrier theorem no longer applies unchanged	$\alpha^{-1}(0) = 137.033\,841\,6$

Table 3: Discrete exclusion structure in the carrier reading. The "no ν^c" row is an external truncation of the hard family packet, not a normal carrier alternative.

In addition, a light seam-even colored triplet zero mode would require a light seam-even bosonic mode in E_3. The minimal carrier picture does not provide such a mode, so the colored partner is not on equal footing with the Higgs candidate in E_2.

8 Relative Spectral Completion Targets

Status: dynamical target, not part of the hard carrier core

The relative spectral action is the natural place to unify the α-, χ-, and metric channels, but the heat-kernel completion is not yet theorem-hard. In v2.9 it should therefore appear as a completion target rather than as part of the rigid carrier packet.

The natural completion ansatz is

$$S_{\rm rel} = {\rm Tr}\, f(D/\chi) - {\rm Tr}\, f(D_{\rm ref}/\chi) + \frac{i\pi}{2}\Delta\eta_\Gamma + S_{\rm defect}.$$

The three stationarity channels are then

$$\frac{\delta S_{\rm rel}}{\delta\alpha} = 0, \qquad \frac{\delta S_{\rm rel}}{\delta\chi} = 0, \qquad \frac{\delta S_{\rm rel}}{\delta e} = 0,$$

with an independent ω variation left to a later Palatini/Einstein–Cartan completion.

8.1 Why the denominator 192 pi squared is plausible

For a Dirac-type Laplace operator written as

$$D^2 = -\nabla^2 + \mathcal{E},$$

the local four-dimensional heat-kernel coefficient is

$$a_2(D^2) = \frac{1}{16\pi^2} \int d^4x\sqrt{-g}\ \mathrm{Tr}\Big(\mathcal{E} + \frac{1}{6}R\Big).$$

For a Dirac-type curvature coupling one has $\mathcal{E} \sim R/4 + \cdots$, so the curvature part scales like

$$\frac{1}{16\pi^2}\left(\frac{1}{4} + \frac{1}{6}\right) \int d^4x\sqrt{-g}\ \mathrm{Tr}(\mathbf{1})\, R = \frac{5}{192\pi^2} \int d^4x\sqrt{-g}\ \mathrm{Tr}(\mathbf{1})\, R.$$

After relative admissible-minus-reference subtraction and TFPT normalization, an Einstein-channel coefficient with denominator $192\pi^2$ is therefore natural. What remains open is the exact numerator and the precise sector trace selected by the full relative operator.

Conjecture 8.1 (Induced Einstein channel)**.** The relative spectral completion should organize the metric channel as

$$S_{\mathrm{EH}}^{\mathrm{rel}} = \frac{\chi^2}{192\pi^2}\ \mathrm{Tr}_{\mathcal{H}_{\mathrm{ch}}}\, \mathbf{1} \int d^4x\sqrt{-g}\, R.$$

On the canonical occupancy branch, where $\mathrm{Tr}_{\mathcal{H}_{\mathrm{ch}}}\, \mathbf{1} = \Omega_{\mathrm{adm}} = 48$, this gives

$$S_{\mathrm{EH}}^{2.9} = \int d^4x\sqrt{-g}\, \frac{\chi^2}{4\pi^2} R = \int d^4x\sqrt{-g}\, \frac{\bar{M}_{\mathrm{Pl}}^2}{2} R,$$

so that

$$\bar{M}_{\mathrm{Pl}}^2 = \frac{\chi_0^2}{2\pi^2}.$$

Using the observed reduced Planck mass only as a calibration check gives

$$\chi_0 = \sqrt{2}\,\pi\, \bar{M}_{\mathrm{Pl}} \approx 1.081\,840\,000 \times 10^{19}\ \mathrm{GeV}.$$

The v2.9 claim is not that this is already hard, but that the relative-a_2 channel is the correct place to make it hard.

9 Downstream Reinterpretations: E_8, Higgs, Flavor

9.1 E_8 as a downstream carrier rereading

The strongest E_8 rereading to retain is

$$\lambda_{E8} = \frac{\gamma}{\ln(248/60)}, \qquad \gamma = \mathrm{Tr}_E\, Y^2 = \frac{5}{6}.$$

The E_8 sector is downstream of the carrier norm, not part of the hard minimal carrier core. One suggestive but non-rigid echo is $60 = 48 + 12 = \Omega_{\mathrm{adm}} + \dim \mathfrak{g}_{\mathrm{SM}}$.[1]

9.2 Higgs as a conditional seam-even mode in E_2

If a light seam-even bosonic zero mode exists in E_2, then the minimal carrier provides a natural Higgs candidate

$$\tau H = +H, \qquad H \in \Gamma(E_2) \cong (1,2)_{1/2}.$$

This is the correct representation-theoretic slot for a Standard Model Higgs doublet. The existence of the light mode itself remains part of the dynamical completion and should not be overstated as a theorem of the hard carrier core.

[1] This identity is numerically suggestive, but not theorem-level and should remain secondary or appendix-level.

9.3 Flavor and the phase grammar

The hard group-theoretic statement is

$$G_{\mathrm{SM}}^{\mathrm{glob}} = S(U(3) \times U(2)) \cong \frac{SU(3) \times SU(2) \times U(1)_Y}{\mathbb{Z}_6}.$$

The separate, still dynamical question is whether the observed CKM, PMNS, and Majorana phases actually realize the associated discrete grammar.

Conjecture 9.1 (Phase grammar)**.** The global quotient data suggest a distinguished six-point phase lattice

$$\delta \in \left\{0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}\right\} \mod 2\pi.$$

With CP acting by inversion,

$$\delta \mapsto -\delta \mod 2\pi,$$

the resulting phase grammar is naturally

$$\mathbb{Z}_6 \rtimes \mathbb{Z}_2.$$

In v2.9 the quotient theorem should therefore be stated hard, while the physical realization of the phase grammar should remain explicitly conjectural.

10 Conclusion

The rigid core of v2.9 is not a number but a theorem:

$$\boxed{\text{admissibility on the orientable double cover and the seam involution}}$$

$$\boxed{\text{force the minimal carrier } E_3 \oplus E_2.}$$

From this follow the global Standard Model group, one chiral family, the carrier norm $\mathrm{Tr}_E\, Y^2 = 5/6$, and the general abelian index formula. Only on the canonical occupancy branch does the same rigid package produce

$$\Omega_{\mathrm{adm}} = N_{\mathrm{fam}} \dim S^+ = 3 \times 16 = 48.$$

The center is therefore the minimal carrier. The number 48 is the first large trace of that carrier packet in an admissible vacuum, not the foundation of the architecture.

A v2.8 to v2.9 Dictionary

v2.8 language	v2.9 carrier reading
$g = 5$	$\gamma = \frac{5}{6}$ together with $B = \frac{3}{2}$ on the minimal carrier packet
48	$\Omega_{\mathrm{adm}} = N_{\mathrm{fam}} \dim S^+$ on the canonical occupancy branch
$b_1 = \frac{41}{10}$	$b_1(N_{\mathrm{fam}}, N_H) = \frac{4}{3} N_{\mathrm{fam}} + \frac{1}{10} N_H$ evaluated at $(3, 1)$
$\delta_2 = \frac{5}{4}\, \delta_{\mathrm{top}}^2$	$\delta_2 = B\gamma\, \Omega_{\mathrm{adm}}^2 c_3^8$ with $B\gamma = \frac{5}{4}$
$\gamma(0) = \frac{5}{6}$ in E_8 language	$\gamma = \mathrm{Tr}_E\, Y^2$ as a carrier invariant

Table 4: Compact dictionary from the v2.8 closure language to the v2.9 carrier language.

B Family-Index Bridge

The v2.8 block relation

$$k_B = \frac{3}{2}\, I_1(B)$$

already shows that family multiplicity is tied to the three distinguished cycles of the orientable double cover. Writing

$$k_B = \frac{1}{2}\sum_{i=1}^{3} I_{\mathrm{fam}}(C_i)\, I_1(B)$$

makes the same factorization explicit and immediately yields

$$\sum_{i=1}^{3} I_{\mathrm{fam}}(C_i) = 3.$$

The equal-one assignment is the minimal symmetric branch and is the natural source of $N_{\mathrm{fam}} = 3$.

C Occupancy Transgression

The canonical bridge is

$$\Omega_{\mathrm{adm}} = N_{\mathrm{fam}} \dim S^+ = 3 \cdot 16 = 48.$$

With $c_3 = 1/(8\pi)$ this gives

$$\Omega_{\mathrm{adm}}\, c_3^4 = 48 \left(\frac{1}{8\pi}\right)^4 = \frac{3}{256\pi^4} = \delta_{\mathrm{top}}.$$

This is the precise sense in which the topological surcharge is re-read as admissible occupancy transgression.

D Seam Descent Normalization

The seam descent logic is intentionally separated into two pieces:

1. the spin-lifted seam loop closes only after 4π, so $\widetilde{U}_\Gamma(2\pi) = -1$ and $\widetilde{U}_\Gamma(4\pi) = 1$;
2. if the admissible carrier modulus is identified with the axionic angle variable, then $a = \theta/(4\pi)$ and therefore $c_3 = 1/(8\pi)$.

The first step is geometric. The second is the explicit carrier-to-axion identification.

E Relative a_2 Sketch

The relative Einstein target rests on the standard four-dimensional heat-kernel structure

$$a_2(D^2) = \frac{1}{16\pi^2}\int d^4x\sqrt{-g}\ \mathrm{Tr}\left(\mathcal{E} + \frac{1}{6}R\right).$$

For Dirac-type curvature coupling $\mathcal{E} \sim R/4 + \cdots$, so curvature terms naturally carry a denominator of the form $192\pi^2$. The open step is not whether such a denominator is plausible, but which relative subtraction and sector trace produce the final canonical coefficient

$$\frac{\chi^2}{192\pi^2}\ \mathrm{Tr}_{\mathcal{H}_{\mathrm{ch}}}\, \mathbf{1}.$$

F Numerical Branch Table

Branch	N_{fam}	N_H	Ω_{adm}	$\alpha^{-1}(0)$
canonical carrier branch	3	1	48	137.035 999 2
boundary multiplicity deformation	2	1	32	156.094 045 5
boundary multiplicity deformation	4	1	64	124.835 054 8
scalar-content deformation	3	2	48	135.945 991 7
external truncation (no ν^c)	3	1	45	137.033 841 6

Table 5: Representative branches in the carrier CFE. The "no ν^c" row is not a normal carrier branch but an external truncation of the hard family packet.